# Task 5:- Cybercrime Awareness Case Study & Prevention Guide

## Topic:- Ransomware

**Ransomware** is a type of cybercrime in which malicious software encrypts a victim's data, rendering it inaccessible until a ransom is paid—usually in cryptocurrency. A well-known real-world example is the **WannaCry ransomware attack**.

In this case, the attack typically unfolds in several stages. First, the attacker distributes the ransomware through phishing emails, malicious downloads, or by exploiting system vulnerabilities. In **WannaCry**, attackers exploited a Windows vulnerability using a leaked tool originally developed by the **National Security Agency**. Second, once inside the system, the malware spreads laterally across networks without user interaction. Third, it encrypts files and displays a ransom note demanding payment in Bitcoin. Finally, victims must decide whether to pay (with no guarantee of recovery) or attempt costly data restoration.

The primary targets are organizations with critical data and limited tolerance for downtime, such as hospitals, universities, and businesses. During the WannaCry incident, the **National Health Service** was severely affected, leading to cancelled appointments and disrupted medical services.

The consequences are significant: financial losses, operational paralysis, reputational damage, and potential legal liabilities. In extreme cases, such as healthcare disruptions, ransomware can indirectly endanger lives. This case highlights the importance of timely software updates, cybersecurity awareness, and robust backup strategies to mitigate such attacks.